FIITJEE

ALL INDIA TEST SERIES

OPEN TEST – II

JEE (Advanced)-2021

PAPER – 1

TEST DATE: 21-03-2021

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*

   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

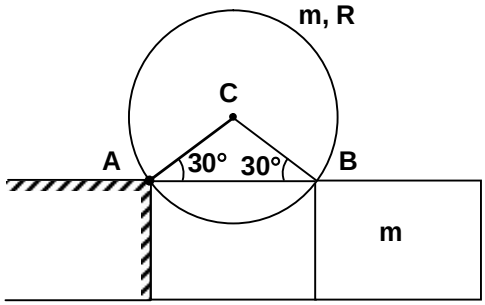
PART – I

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

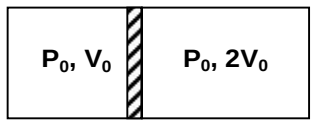
1. A particle moves along a circle of radius R = 10m with retardation so that at any time the modulus of tangential acceleration is twice the modulus of normal acceleration. If initial speed of the particle is v_0 = 20m/s. Then the speed of the particle after having covered a distance s = 10m would be (take e^{-2} = 0.135)
   (A) 1.7 m/s
   (B) 2.7 m/s
   (C) 3.7 m/s
   (D) 4.7 m/s

2. A uniform circular ring of mass m = 5 kg and radius R = 30 cm is hinged at the point 'A' of a stationary surface with the line joining point 'A' to the centre 'C' of the ring is initially vertical. The ring is also in contact with the Plank of mass m = 5 kg at point 'B' which is placed on a smooth horizontal surface. Initially points 'A' and 'B' are almost touching and the line AC is vertical and the system is released from rest. Then the velocity of the Plank when the line AC makes an angle 30° with the horizontal as shown will be (there is no friction between the ring and the plank and g = 10 m/s^2)



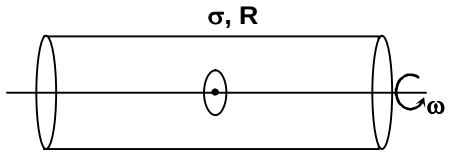
   (A) 1 m/s
   (B) 2 m/s
   (C) 3 m/s
   (D) 4 m/s

3. A horizontal cylinder is divided into two chambers by a thin smooth heat insulating piston of mass 'm' and cross section area 'A' initially. When the piston is in equilibrium, pressure of gas in each chamber is P_0 and volumes of the chambers are V_0 and $2V_0$ as shown in the figure. The ideal gases in the two chambers are kept at constant temperatures T_1 and T_2. The piston is slightly displaced from equilibrium position and then released. Then the time period of small oscillations of the piston is



   (A) $2\pi\sqrt{\frac{mv_0}{3P_0A^2}}$
   (B) $2\pi\sqrt{\frac{2mv_0}{3P_0A^2}}$
   (C) $2\pi\sqrt{\frac{3mv_0}{2P_0A^2}}$
   (D) $2\pi\sqrt{\frac{mv_0}{2P_0A^2}}$

4. The curved surface of a very long hollow non conducting cylinder of radius 'R' is uniformally charged with surface charge density 'σ'. A non conducting small circular ring of radius 'a' and mass 'm' having charge 'q' uniformly distributed over its length is placed coaxially inside the hollow cylinder at its centre. The arrangement is located in a gravity free space.



If the cylinder is rotated with a constant angular velocity 'ω' about its axis as shown in the figure. Then the angular velocity acquired by the ring is

(A) $\frac{4\mu_0 q\sigma\omega R}{m}$

(B) $\frac{2\mu_0 q\sigma\omega R}{m}$

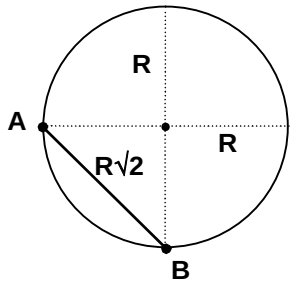
(C) $\frac{\mu_0 q\sigma\omega R}{m}$

(D) $\frac{\mu_0 q\sigma\omega R}{2m}$

(One or More than one correct type)

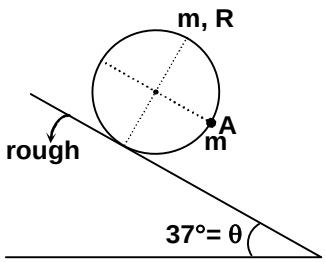
This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. Two small balls 'A' and 'B' of masses 0.3 kg and 0.6 kg respectively are rigidly attached to the ends of a light rigid rod of length $R\sqrt{2}$ and placed inside a fixed smooth spherical cavity of radius R = 0.5 m with the ball 'B' is located at the lowest point of cavity as shown in the figure. Then choose the correct option(s) immediately after the rod is released from rest. (Take g = 10 m/s^2)



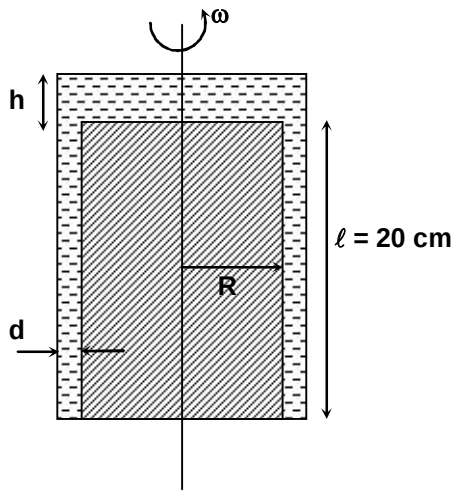
(A) The normal force on the ball 'A' due to surface of cavity is 2N.
(B) The normal force on the ball 'A' due to surface of cavity is 4N.
(C) The normal force on the ball 'B' due to surface of cavity is 8N.
(D) The force acting on the ball 'B' due to light rod is $2\sqrt{2}$ N.

6. A particle of mass m = 1 kg is rigidly attached to the circumference at point 'A' of a uniform circular ring of mass m = 1kg and radius R = 20 cm placed on a rough inclined plane of inclination θ = 37° as shown in the figure. Then choose the correct option(s) just after the ring is released from rest. (there is no slipping between the ring and inclined plane and g = 10 m/s^2)



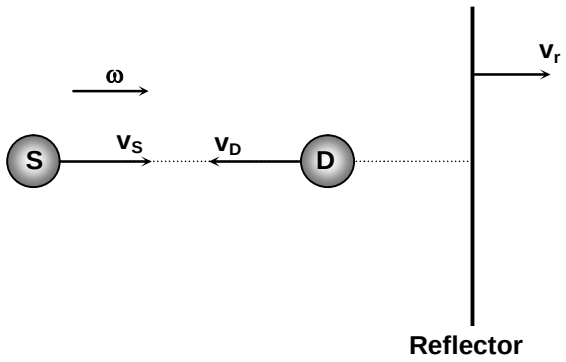
(A) The frictional force acting on the ring due to inclined plane is 4N.
(B) The frictional force acting on the ring due to inclined plane is 2N.
(C) The normal force acting on the ring due to inclined plane is 22 N.
(D) The normal force acting on the ring due to inclined plane is 11N.

7. A uniform solid cylinder of radius R = 10 cm and length ℓ = 20 cm is placed coaxially inside a stationary cylindrical cavity. The gap between the cylinder and the cavity is filled with the viscous liquid of viscosity η = 0.07 poise as shown in the figure. The cylinder is rotated with a constant angular velocity ω = 50 rad/s about its vertical axis. Then choose the correct option(s). (Given h = 1 mm, d = 0.5 mm)



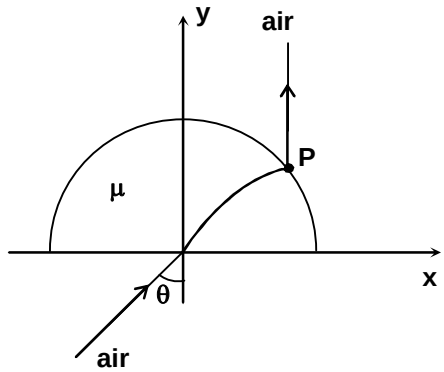
(A) The net torque of viscous force acting on the cylinder is 0.935 N-m.
(B) The net torque of viscous force acting on the cylinder is 0.880 N-m.
(C) The power developed due to viscous force acting on the cylinder is 42.50 Watt.
(D) The power developed due to viscous force acting on the cylinder is 46.75 Watt.

8. In the arrangement shown in the figure, a reflector is moving towards right with a velocity v_r = 20 m/s. Source and detector are moving towards each other with the velocities v_S = 30 m/s and v_D = 10 m/s respectively. The wind is blowing with a velocity ω = 10 m/s towards the reflector. The frequency of sound emitted by the source is f = 527 Hz and the velocity of sound with respect to air is v = 330 m/s. Then choose the correct option(s).



(A) The frequency of the reflected wave received by the detector is 595 Hz
(B) The frequency of the reflected wave received by the detector is 496 Hz
(C) The wavelength of the reflected wave received by the detector is $\left(\frac{5}{8}\right)$ meter.
(D) The wavelength of the reflected wave received by the detector is $\left(\frac{10}{17}\right)$ meter.

9. A hemisphere of radius $\frac{a}{2}$ and made up of a material of variable refractive index is placed with its base centre 'O' at the origin as shown in the figure. The refractive index of the material of the hemisphere varies as $\mu = \left(\frac{a}{a-x}\right)$. A ray of light is incident at the point 'O' at an angle θ(=0) with the normal in the x-y plane and it comes out through a point 'P' on its curved surface. Then choose the correct option(s).



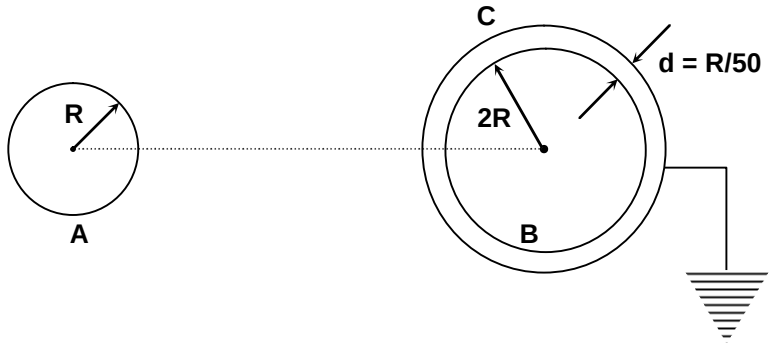
(A) The equation of trajectory of ray of light inside the hemisphere is $y = \sqrt{2ax - x^2}$

(B) The equation of trajectory of ray of light inside the hemisphere is $y = \sqrt{ax - 4x^2}$

(C) The x and y coordinates of point 'P' are $\left(\frac{a}{8}, \frac{a}{8}\sqrt{15}\right)$

(D) The x and y coordinates of point 'P' are $\left(\frac{a}{2}, \frac{a}{2}\sqrt{15}\right)$

10. Two conducting spheres A and B of radius R and 2R are connected by a thin conducting wire. The conducting sphere 'B' is surrounded by a grounded thin concentric conducting shell 'C' with radius $\left(\frac{101}{50}R\right)$. The connecting wire between the spheres A and B is not touching the thin shell 'C'. If conducting sphere 'B' is given a charge 'Q' and the separation between the spheres A and B is much larger than their radii. Then choose the correct option(s).



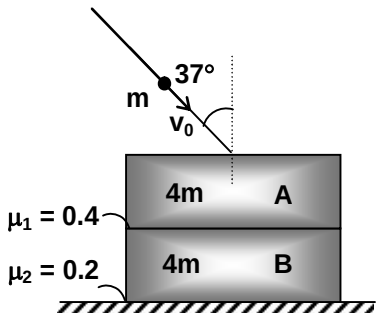
(A) The charge appearing on the conducting sphere 'A' is $\frac{Q}{201}$.

(B) The charge appearing on the conducting sphere 'A' is $\frac{Q}{101}$.

(C) The charge appearing on the conducting sphere 'B' is $\frac{100Q}{101}$.

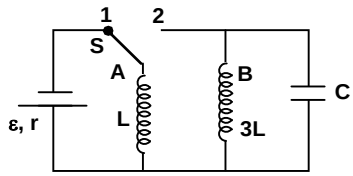
(D) The charge appearing on the conducting sphere 'B' is $\frac{200Q}{201}$.

11. A block 'A' of mass 4m is placed on the block 'B' of same mass 4m which is placed on a rough horizontal surface with coefficient of friction $\mu_2 = 0.2$. The coefficient of friction between the blocks is $\mu_1 = 0.4$. A ball of mass m moving with a velocity $v_0 = 10$ m/s collides with the block 'A' at an angle $\theta = 37°$ with the normal to the surface as shown and gets stuck to it. Then choose the correct option(s).



(A) The velocity of block 'A' just after collision is 0.56 m/s.
(B) The velocity of block 'A' just after collision is 0.28 m/s.
(C) The velocity of block 'B' just after collision is 0.40 m/s.
(D) The velocity of block 'B' just after collision is 0.20 m/s.

12. A circuit consists of two inductors 'A' and 'B' of inductances L and 3L respectively and capacitor of capacitance 'C' as shown in the figure. The emf of battery is ε and its internal resistance 'r'. Initially the switch was in position '1' for a long time and then it is shifted to position '2'. Then choose the correct option(s).



(A) The maximum charge on the capacitor is $\frac{\varepsilon}{r}\sqrt{3LC}$

(B) The maximum charge on the capacitor is $\frac{\varepsilon}{2r}\sqrt{3LC}$

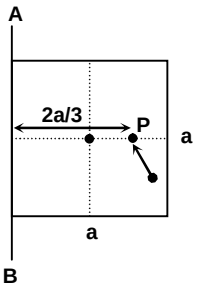
(C) The maximum current through the inductor 'B' is $\frac{\varepsilon}{2r}$

(D) The maximum current through the inductor 'B' is $\frac{3\varepsilon}{4r}$

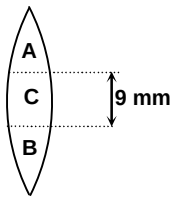
SECTION – C
(Numerical Answer Type)

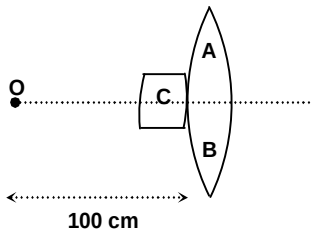
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A thin uniform square plate of side a = 1m and mass M = 2kg can freely rotate about a stationary vertical axis AB coinciding with one of its sides. A small ball of mass m = 0.5 kg moving with a horizontal velocity v_0 = 6 m/s at right angle to the plate collides elastically at point 'P' of the plate which is located at a distance $\frac{2a}{3}$ from the vertical axis AB as shown in the figure. Find the horizontal component of the resultant force (in Newton) that will be exerted by the axis AB on the plate after collision.

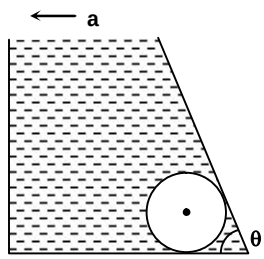


14. A thin convex lens of focal length 50 cm is cut into three parts A, B and C by the planes parallel to its optic axis. The thickness of the middle layer C is 9 mm. The middle layer is now removed and the two parts A and B are stuck together. Then the part C is also placed in contact of this lens symmetrically as shown in the figure. Now, a point object 'O' is placed at a distance 100 cm left on the optic axis of the part C. Find the distance (in mm) between the two images formed by the parts A and B.

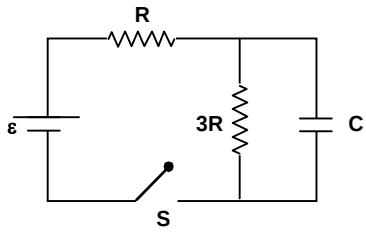




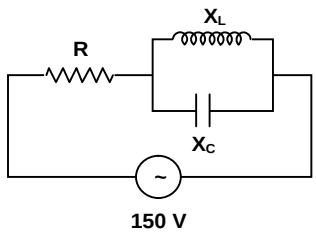
15. A glass sphere of volume V = 500 cm^3 is placed in a vessel filled with water A side wall of the vessel is inclined at an angle θ = 37° with the horizontal bottom of the vessel. The container is moving with a uniform leftward acceleration a = 6 m/s^2. The density of the glass is ρ = 2500 kg/m^3 and the density of water is ρ_w = 1000 kg/m^3 and g = 10 m/s^2. Find the normal force (in newton) between the bottom of the vessel and the glass sphere.



16. In the circuit shown, the power dissipated in the resistances R and 3R at an instant after the switch is closed are 16 W and 3W respectively. Find the rate of increase in the energy stored in the capacitor at this instant (in watt).



17. In the ac circuit shown in the figure, X_L = 8 Ω, X_C= 4 Ω and R = 6Ω. The rms voltage of ac source is 150 V. Find the rms current (in ampere) through the inductor.



18. Consider the following nuclear fusion reaction

$${}_1^2H + {}_1^2H \rightarrow {}_1^3H + {}_1^1P$$

Assume binding energy per nucleon of deuterium and tritium are 1 MeV and 2.80 MeV respectively. Neglect the kinetic energy of the nuclei before the fusion. Find the kinetic energy (in MeV) of the tritium produced in the above nuclear reaction.

Chemistry

PART – II

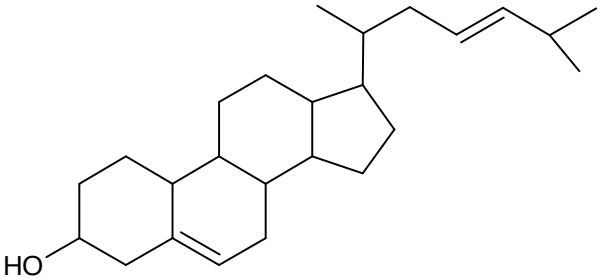
SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Identify the compound which differ in its colour from rest of the three:
    (A) Potassium hexanitritocobaltate(III)
    (B) Tetraamminecopper(II) sulphate
    (C) Ferric ferrocyanide
    (D) Hexaamminenickel(II) chloride

20. Identify the only 'INCORRECT' statement among the following:
    (A) In general, all stereospecific reactions are stereoselective but the reverse is not necessarily true.
    (B) A free radical can be generated at bridge head position.
    (C) During hydrogenation of olefins by Wilkinson's catalyst conjugated double bond is preferred over a non-conjugated double bond.
    (D) Singlet carbene $\left(:CCl_2\right)$ is more stable than triplet carbene $\left(:CCl_2\right)$

21. Consider the following molecule



The number of stereocentres and stereoisomers respectively of the above molecule are:
    (A) 8, 2^8
    (B) 9, 2^9
    (C) 10, 2^{10}
    (D) 10, 2^9

22. In a sample of H-atoms, all atoms are initially in their 4th excited state and now these atoms are allowed to de-excite to the ground state. Now, choose the only INCORRCT statement of the following: (Assume multiple excitation of atoms is not allowed)
    (A) If the sample contains only two H-atoms, then a maximum of six distinct spectral lines can be observed in its emissions spectrum.
    (B) Minimum five hydrogen atoms, the sample should contain in order to show all possible spectral lines in its emission spectrum
    (C) If the sample contains 8 H-atoms, then maximum number of 10 spectral lines can be observed, in its emission spectrum
    (D) If the sample contains minimum 6 H-atoms, then a maximum of 10 spectral lines can be observed in its emission spectrum.

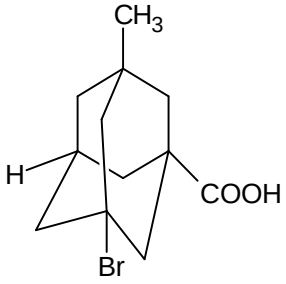
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

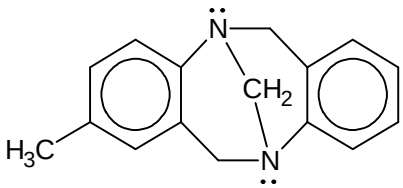
23. During an adiabatic reversible expansion, a gas obeys VT^3 = constant. The gas may be:
(A) O_2
(B) O_3
(C) He
(D) SO_2

24. Following two equilibria are established on mixing two gases A_2 and C.
(i) $3A_2(g) \rightleftharpoons A_6(g), K_p = 1.6\ atm^{-2}$
(ii) $A_2(g) + C(g) \rightleftharpoons A_2C(g)$
A_2 and C are mixed in 2 : 1 molar ratio in seal container at a constant temperature. If the total pressure of the reaction mixture is given to be 1.4 atm and partial pressure of A_6 to be 0.2 atm at equilibrium, then identify the correct statement(s) of the following:
(A) Equilibrium partial pressure of A_2 (g) is 0.5 atm.
(B) Equilibrium partial pressure of 'C' is 0.4 atm
(C) Equilbirium partial pressure of A_2C (g) is 0.3 atm
(D) K_P of reaction (ii) is 1.5 atm^{-1}

25. Consider the structure of two compounds P and Q:



(P)



(Q)

Which of the following statement(s) is/are correct regarding P and Q?
(A) P is chiral
(B) Q is chiral
(C) In 'Q' nitrogen inversion is prevented
(D) 'P' exists as two pair of enantiomers

26. A mixture containing acetone and its next two homologues are heated with hydroxyl amine in presence of catalytic amount of sulphuric acid. Choose the correct statement(s) regarding this reaction from the options given below:
(A) It is a nucleophilic addition reaction.
(B) It is a condensation reaction.
(C) Attack of H^+ ion is the slow step in this reaction.
(D) A mixture of eight oximes will be produced in this reaction.

27. A transient blue/violet colouration may be obtained by mixing

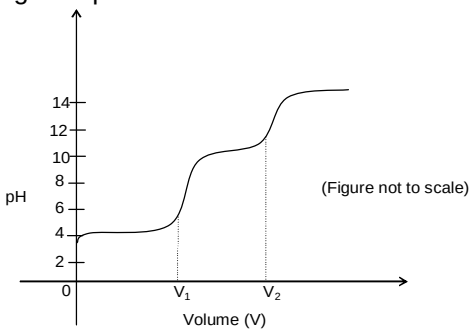
(A) $Na_2S_2O_3$ and Fe^{3+} (acidified)

(B) $Na_2S_2O_3$ and Fe^{3+} (neutral)

(C) KI + F_2

(D) $FeSO_4 + K_3\left[Fe(CN)_6\right]$

28. A weak acid is titrated against a strong base and the volume (V) of titrant added, is plotted against pH as shown below:



Which of the following statement(s) regarding above titration is/are correct?

(A) The substance in beaker is a solution of a weak dibasic acid.

(B) $V_2 = 2V_1$.

(C) The pH corresponding to $\left(\frac{V_1}{2}\right)$ corresponds to pK_{a_1} of the weak dibasic acid being titrated.

(D) The pH corresponding to $\left(\frac{3V_1}{2}\right)$ corresponds to pK_{a_2} of the weak dibasic acid to be titrated.

29. $[X] + H_2O_2 + H^+ \longrightarrow \underset{(\text{Blue colour})}{[Y]} + H_2O$

$[Y] + H^+ \longrightarrow \underset{(\text{Green soln})}{[Z]} + M\uparrow + H_2O$

Now, identify the correct statement(s) regarding above sequence of reactions:

(A) 'X' may be K_2CrO_4 or $K_2Cr_2O_7$.

(B) 'Y' is CrO_3.

(C) Green colour of (Z) is due to the presence of Cr^{3+}.

(D) Gas (M) is O_2.

30. A crystalline cubic solid is made up of atoms A, B, C and D such that atoms 'A' are present at each of the corner, 'B' atoms at each of face centre, 'C' atoms in each of octahedral void and 'D' atoms in each of the tetrahedral voids of the unit cell. Now choose the correct statement(s) regarding the above solid.
(Atoms from the lattice sites of the unit cell have to removed completely and not partially)

(A) If all atoms lying along or inside one of the octant are removed, then the formula of resulting solid would be $A_7B_{16}C_{12}D_{56}$.

(B) If all atoms lying along one of the C_4 – axis of symmetry are removed, then the formula of the resulting solid would be $AB_2C_3D_8$.

(C) If all atoms lying along one of the C_3 – axis of symmetry are removed, then the formula of resulting solid would be $AB_4C_4D_8$.

(D) If all the atoms lying along one of the C_2 – axis of symmetry are removed, then the formula of resulting solid would be $A_2B_6C_5D_{16}$.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

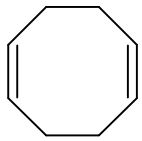
31. The collision frequency of a sample of an ideal gas molecules is defined as the ratio of the average speed of gas molecule to their mean free path. So, if the pressure and temperature of the gas molecules is made 8 times of its initial pressure and temperature, then ratio of new collision frequency to that of original collision frequency will be (given $\sqrt{2} = 1.41$, $\sqrt{3} = 1.73$)

32. The mechanism of the reaction, $A \longrightarrow C$ is given below:

$$A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B(\text{fast})$$

$$B \xrightarrow[K_2]{} C(\text{slow})$$

By using steady state approximation, the rate law of the above reaction is given by
$r = k[A]$
Where k is the rate constant of the overall reaction $(A \longrightarrow C)$.
So, if $k_f = 2\ \text{sec}^{-1}$, $k_b = 0.4\ \text{sec}^{-1}$, and $k_2 = 0.1\ \text{sec}^{-1}$, the value of 'k' will be (in units of sec^{-1})

33. 

$$\xrightarrow[(1\,\text{eqvt})]{BH_3-THF} \underset{(\text{Major})}{(B)} \xrightarrow{\text{cis}-4-\text{methyl 2}-\text{pentene}} \underset{(\approx 100\%)}{(C)}$$

(A)

Now, let, the degree of unsaturation of (B) = x
the degree of unsaturation of (C) = y
Total number of stereoisomers of (B) = z
Total number of stereoisomers of (C) = m
Then, calculate the value of $\dfrac{(xy+1)}{(m+z)}$.

34. Iodine reacts with ozone gas to form a dark yellow solid (X). Let, the number of lone – pair of electrons in un-ionised form of (X) be m, number of lone pair of electrons in the anionic moiety of (X) be 'n' and the positive charge on the cationic moiety of (X) be 'p' units. Then the value of $\dfrac{(m-p)}{n}$ is.....

35. The ratio of the sum of number of resonating structures and number of plane of symmetry in the anionic moiety of gypsum to the number of resonating structures in the anionic moiety of calcium phosphate is...

36. Ferric sulphide reacts with oxygen gas to produce ferrous sulphate and sulphur dioxide. The gram equivalent weight of ferric sulphide in the above reaction is:
(Atomic weight : Fe = 56, S = 32)

Mathematics PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If $z = x + iy$, $x, y \in R$, $i = \sqrt{-1}$, such that $|z| = 1$, then the maximum value of $|z^3 - z + 2|$ is?
(A) 2
(B) 4
(C) $\sqrt{17}$
(D) $\sqrt{13}$

38. If $E_1, E_2, E_3, \ldots, E_{12}$ are distinct numbered edges of a cube, then the number of unordered pairs (E_i, E_j) where $1 \le i, j \le 12$ that determine a plane is?
(A) 38
(B) 42
(C) 36
(D) 12

39. If $1+\frac{p}{q} = \sum_{i=0}^{1000} \frac{{}^{1000}C_i}{{}^{2020}C_i}$, where p, q are coprime, then p + q is equal to?
(A) 3020
(B) 1510
(C) 2021
(D) 2019

40. If $I = \int_0^{\pi/4} \frac{\tan^2 x}{x^2 + \tan^2 x - 2x\tan x - 3x + 3\tan x + 2} dx$, then value of I is?
(A) $\frac{1}{2}\ln 2$
(B) $-\ln\left(\frac{16-2\pi}{12-\pi}\right)$
(C) $\ln\left(\frac{16-2\pi}{12-\pi}\right)$
(D) $-\frac{1}{2}\ln 2$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. If PQRS is a tetrahedron such that $\overrightarrow{PQ} = \vec{p}$, $\overrightarrow{PR} = \vec{q}$, $\overrightarrow{PS} = \vec{r}$, $|\vec{p}| = \sqrt{3}$, $|\vec{q}| = \sqrt{2}$, $|\vec{r}| = \sqrt{5}$, $\vec{p}\cdot\vec{q} = 2$, $\vec{q}\cdot\vec{r} = -1$ and $\vec{r}\cdot\vec{p} = -1$, then?
(A) volume of tetrahedron is 1 unit^3
(B) length of altitude from vertex S to the opposite face is $\frac{3}{\sqrt{2}}$ unit

(C) volume of tetrahedron is $\frac{1}{2}$ unit^3

(D) length of altitude from vertex S to the opposite face is $3\sqrt{2}$ unit

42. Let $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, and S denote the set of all the complex numbers in the Argand plane of the form $a + b\omega + c\omega^2$, where a, b, c $\in$ [0, 1]?

(A) length of perimeter of region traced by S is $\sqrt{3}\,\pi$ unit

(B) area of region traced by S is $\frac{3\sqrt{3}}{2}$ unit^2

(C) length of perimeter of region traced by S is 6 unit

(D) area of region traced by S is $\frac{\pi\sqrt{3}}{2}$ unit^2

43. If A_n is the number of sequences of 0's and 1's of length n that begin with a 0, end with a 0, contain no two consecutive 0's and contain no three consecutive 1's, then?

(A) $A_5 = 1$

(B) $A_6 = 2$

(C) $A_{10} = A_8 + A_9$

(D) $A_{11} = A_8 + A_9$

44. There are 17 identical red balls and 10 identical white balls be distributed among 4 distinct boxes, then which of the following is/are CORRECT?

(A) number of ways such that number of red balls is greater then number of white balls in each box is ${}^6C_3 \cdot {}^{13}C_3$

(B) number of way such that number of red balls is greater than the number of white balls in each box is ${}^6C_2 \cdot {}^{20}C_3$

(C) number of ways such that exactly 2 boxes have red balls is ${}^4C_2 \times {}^{16}C_1 \times {}^{12}C_2$

(D) number of ways such that exactly 3 boxes have white balls and each box has different number of white balls is ${}^4C_3 \times 3! \times {}^{20}C_3 \times {}^4C_1$

45. Let a_n and b_n be the sequence of real numbers such that $(2 + i)^n = a_n + b_n i$, for all integers $n \geq 0$ where $i = \sqrt{-1}$, then?

(A) $\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \frac{7}{16}$

(B) $\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \frac{1}{20}$

(C) $\sum_{n=0}^{\infty} \frac{a_n^2 - b_n^2}{7^n} = -\frac{1}{20}$

(D) $\sum_{n=0}^{\infty} \frac{a_n^2 - b_n^2}{7^n} = \frac{7}{8}$

46. Let $S_k = \alpha^k + \beta^k + \gamma^k$, where α, β, γ are the roots of the equation $x^3 - 5x^2 + 8x - 13 = 0$, then?

(A) $S_0 + S_1 + S_2 = 17$

(B) $S_0 - S_1 + S_2 = 7$

(C) if $S_4 = aS_3 + bS_2 + cS_1$, then $a + b + c = 10$

(D) if $S_4 = aS_3 + bS_2 + cS_1$, then $a - b + c = 10$

47. If $P(x) = x^5 + ax^4 + bx^3 + cx^2 + dx - 32$ (where a, b, c d $\in$ R) has the property that whenever r is a root of P(x) so is $r\omega$ (where $\omega = e^{i\frac{2\pi}{3}}$), then?
(A) the value of a can be –4
(B) the value of b can be 4
(C) the value of a can be 2
(D) the value of P(2) can be –7

48. Given a nine-sided regular polygon whose vertices are $A_1\ A_2\ A_3\ A_4$ ..... A_9. Let S = {A_1, A_2, ..., A_9}, then which of the following is/are correct?
(A) Number of distinct isosceles triangles, all of whose vertices belong to the set S is 30
(B) Number of distinct equilateral triangles, at least two of whose vertices belong to the set S is 72
(C) Number of distinct isosceles triangles, all of whose vertices belong to the set S is 33
(D) Number of distinct equilateral triangles, at least two of whose vertices belong to the set S is 66

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. The value of $600\sum_{a=1}^{\infty}\sum_{b=1}^{\infty}\sum_{c=1}^{\infty}\frac{ab(3a+c)}{4^{a+b+c}(a+b)(b+c)(c+a)}$ is?

50. Let a, b, c $\in$ R and $a^2 + b^2 + c^2 = 1$, then the maximum value of |a – b| + |b – c| + |c – a| is?

51. Let $z = \frac{1+i}{\sqrt{2}}$, then the value of $\left(z^{1^2} + z^{2^2} + z^{3^2} + ... + z^{12^2}\right)\left(\frac{1}{z^{1^2}} + \frac{1}{z^{2^2}} + ... + \frac{1}{z^{12^2}}\right)$ is equal to?

52. The area of region traced by the point P(x, y) in Cartesian plane which satisfies the inequation $\sin^2 \pi x + \sin^2 \pi y > 1$, if $x, y \in [-1, 1]$ is?

53. Let f(x) be a third degree polynomial with real coefficients satisfying the equation |f(1)| = |f(2)| = |f(3)| = |f(5)| = |f(6)| = |f(7)| = 12, then the value of |f(0)| is?

54. If $P(n) = \prod_{r=3}^{n}\frac{(r^3+3r)^2}{r^6-64}$, then value of $\lim_{n\to\infty} P(n)$ is, (n $\in$ N)?